

Phases: Augmenting a Live Audio-Visual Performance with AR

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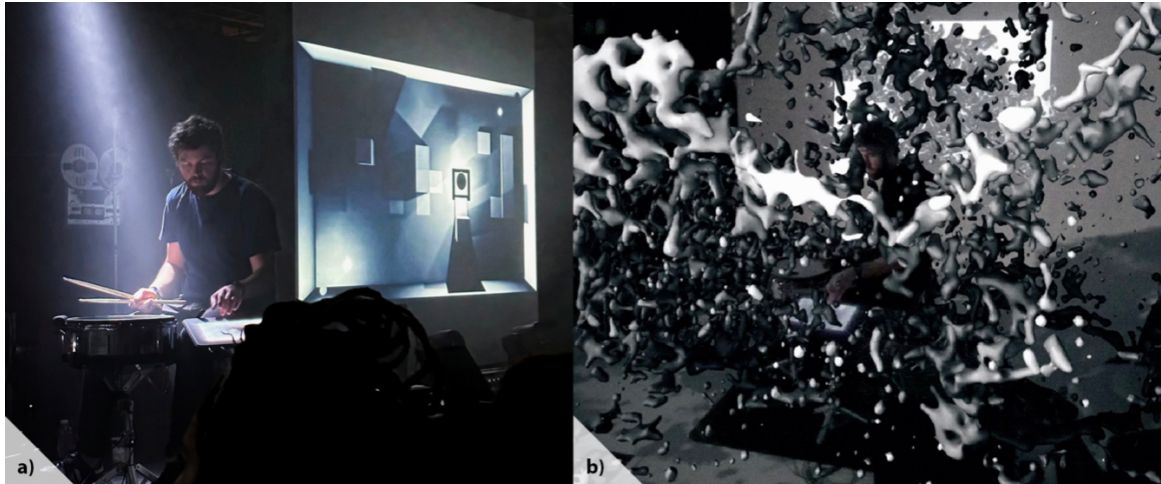


Figure 1: Image a) shows Brett Bolton while performing his audiovisual act at MUTEK Montreal, 2023. Image b) shows the same performance extended with Augmented Reality (AR) visuals around the performer, experienced through AR glasses.

ABSTRACT

Our performance combines stage visuals with Augmented Reality (AR) to demonstrate a novel way of experiencing live events. Our goal is to enhance the sociability of such events by showcasing the potential of AR glasses over smartphone technology, which forces the audience to look at their phones and isolates them from their environment. Our project explores how the latest AR and visual technology enhance the spectator experience while retaining the social and inclusive environment of live performances.

KEYWORDS

Augmented Reality, Real-Time Rendering, Audio-Visual

ACM Reference Format:

Dávid Maruscsák, Brett Bolton, Bent Stamnes, Matt Swoboda, and Christian Sandor. 2024. Phases: Augmenting a Live Audio-Visual Performance with AR. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Real-Time Live! '24*, July 27–August 01, 2024, Denver, CO, USA.

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SIGGRAPH Real-Time Live! '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0526-7/24/07

<https://doi.org/10.1145/3641520.3665310>

01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641520.3665310>

1 INTRODUCTION

Recently, it has become nearly impossible to enjoy a concert without struggling to see the stage through a sea of smartphones. Technology such as smartphones and social media has arguably diminished the live experience, reducing sociability and immersion for those watching through their phones and disrupting the experience for others. This behaviour sacrifices the shared experience of the moment in exchange for a poor video recording or photo, usually intended for posting on social media. The question arises: why be "here" at all? AR glasses have the potential to change this: liberating the viewer from observing through the smartphone and enhancing the immersion of the experience in the moment, in a completely live way that must be experienced in person, in the moment, and retaining - even enhancing - the sociability and shared experience. These are important aspects to explore for the future of live entertainment, and this unique collaboration across industry, academia and creativity offers a unique opportunity to do so.



Figure 2: The image sequence shows the evolution of the visuals starting from left to right from different points of view.

2 TECHNICAL OVERVIEW

Our work is a live, real-time audio-visual performance to be experienced simultaneously via a large screen and through AR glasses. To realise this, we focus on a number of key elements:

Real-time generation and rendering of live visuals, triggered by a performer and created in the latest version of Notch. [10Bit FX Limited, 2024] Notch provides a graphical user-interface (UI) allowing artists to create compelling visuals that are responsive to a range of live inputs, controllers and sensors, and is already used throughout the live entertainment industry by many of the world’s biggest artists, bands and live TV productions. Notch’s new rendering engine is technically innovative in the real-time space, providing a hybrid rasterisation and ray-tracing renderer (based on RESTIR [Ouyang et al., 2021]), particle fluids based on MPM and a sparse volumetric 3D fluid solver. In our performance, multi-disciplinary artist Brett Bolton [Bolton, 2023] showcases these capabilities with visuals that react to his live playing of an electronic drum and MIDI controller.

On-stage two Canon MREAL X1 [Leonard, 2022] AR glasses deliver the experience. AR can superimpose elements onto the environment and alter the perception of the real world itself. Where Virtual Reality (VR) is an isolated experience, AR has the potential to be social - enhancing the experience for viewers sharing the physical environment. Our focus in this area is both technical: aligning scene reconstruction with the coordinate system of the AR glasses and real-time rendering of the visual; and creatively exploratory: what kind of AR experience works within a shared space with non-AR viewers, adds tangible value and retains sociability?

3 EXPERIENCE

We showcase the live audio-visual performance by multi-disciplinary artist Brett Bolton, who drives real-time visuals via his musical performance using custom instruments. While the SIGGRAPH Real-Time Live audience experiences the visuals via the main screen, two audience members will stand on the stage, wearing AR glasses - and experience an enhanced, augmented, immersive view of the performance in real-time.

We aim to demonstrate that the performance can be compelling both with and without AR, simultaneously and together in the same physical space, with the AR experience offering tangible enhancement and increased engagement but with inclusion, not isolation.

The show starts with subtle AR elements that are becoming more and more prominent by the end. The AR elements consist of gas and liquid simulations occupying the space around the performer. The other part of the AR element is taking the video feed of the AR glass as input to create alterations to the real world, altering the viewers’ perception. The third type of AR visuals is facilitating the interaction between the spectators. These visuals are interactive, and the gaze of the spectators triggers them. For example, when the two spectators’ gaze meet, it triggers an action which visually shows a connection between them.

ACKNOWLEDGMENTS

This work was supported by a French government grant managed by the Agence Nationale de la Recherche as part of the France 2030 programme, reference ANR-22-EXEN-0002 (eNSEMBLE/CATS) and by Canon Inc.

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